

Time Series Analysis 2026 Capstone Project

Predictive Modelling and Portfolio Allocation via StockGro

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1. Methodology and Models Used

The objective of this capstone project was to construct a data-driven, risk-optimized ₹10,00,000 virtual portfolio on the StockGro platform. To achieve this, a comprehensive machine learning pipeline was developed using historical daily stock data from the National Stock Exchange (NSE).

To ensure robust predictions, an object-oriented forecasting architecture was built utilizing three distinct models:

- **SARIMA (Seasonal Autoregressive Integrated Moving Average):** Utilized to capture strict seasonal trends (configured for a 5-day weekly trading cycle).
- **Prophet:** Deployed to identify overarching macroeconomic trends and handle structural shifts in the time series.
- **LSTM (Long Short-Term Memory):** A deep learning neural network implemented to capture complex, non-linear market patterns and sudden high-volatility price shocks.

Risk management was executed via mathematical portfolio optimization, specifically utilizing GARCH(1,1) for volatility estimation and the Minimum Variance (Markowitz) strategy for final capital allocation.

2. Stock Selection Rationale (Task 1)

To mitigate sector-specific risk and ensure a well-diversified portfolio, five distinctly unique large-cap stocks were selected from the NSE. This diversification strategy ensures that a downturn in one specific industry is counterbalanced by stability or growth in another.

- **TCS (IT Sector):** Selected as a defensive anchor due to its historical stability, lower daily variance, and consistent long-term trend lines.
- **State Bank of India - SBIN (Banking):** Represents the financial backbone of the economy, sensitive to interest rate changes but highly liquid.
- **Mahindra & Mahindra - M&M (Auto):** Captured for its momentum in both the traditional automotive and emerging EV sectors.
- **Hindustan Aeronautics Limited - HAL (Defense):** Selected as a high-growth, high-volatility asset heavily driven by government contracts and momentum trading.
- **Titan Company - TITAN (Consumer Durables):** Added to capture cyclical consumer spending and seasonal demand spikes.

3. Data Preprocessing Steps (Task 2)

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High-quality forecasting requires a deep understanding of the underlying data distribution. Historical daily price data was fetched via the yfinance API for the period covering January 1, 2021, to December 31, 2025. The preprocessing pipeline included the following rigorous steps:

- **Data Cleaning:** Missing values (NaNs) caused by market holidays or API fetch errors were handled using forward-fill techniques to prevent data leakage and maintain the chronological integrity of the time series.
- **Exploratory Data Analysis & Stationarity Testing:** The Augmented Dickey-Fuller (ADF) test was conducted on the raw 'Close' prices. The results confirmed that the financial time series was inherently non-stationary (exhibiting clear stochastic trends).
- **Model-Specific Data Pipeline Routing:** Although stationary transformations (such as differencing) were explored during EDA, the raw, non-stationary 'Close' prices were deliberately preserved for the final model training pipeline. This strategic decision was made because:
 1. The **Prophet** algorithm requires raw data to natively fit its trend and changepoint detection components.
 2. The **SARIMA** algorithm's integration parameter (d) automatically handles the required differencing internally.
 3. The **LSTM** network utilizes localized scaling rather than global differencing to map non-linear price trajectories.
- **Train/Test Split:** The dataset was sequentially split into a Training set and a Testing set to evaluate out-of-sample accuracy before generating future forecasts.

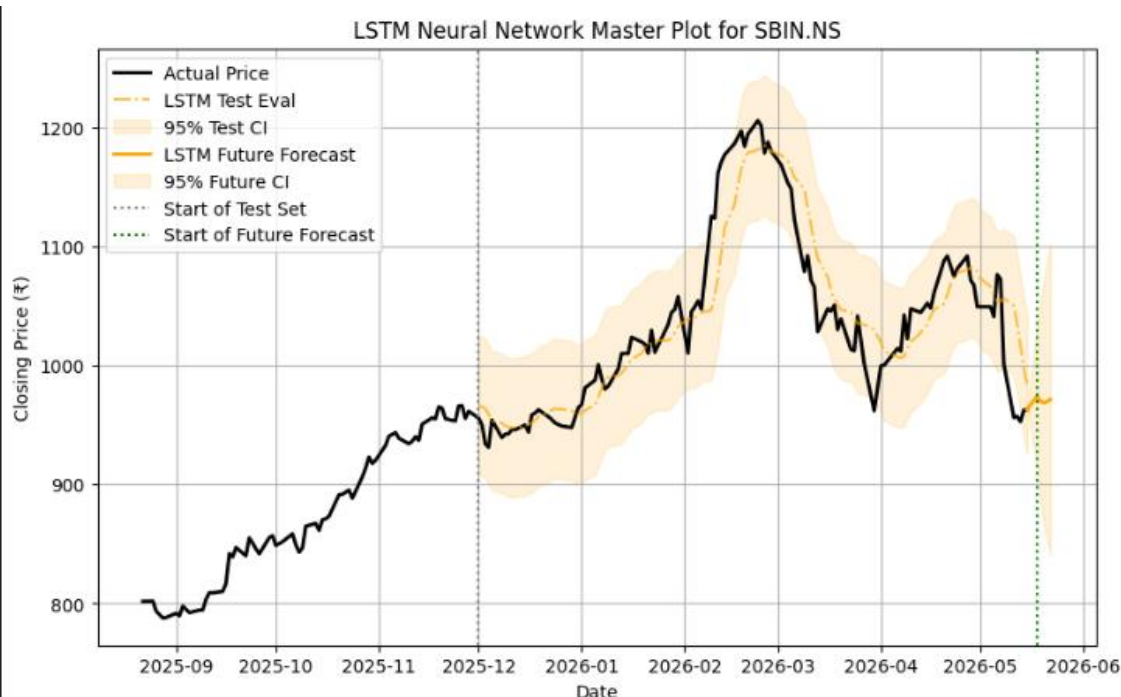
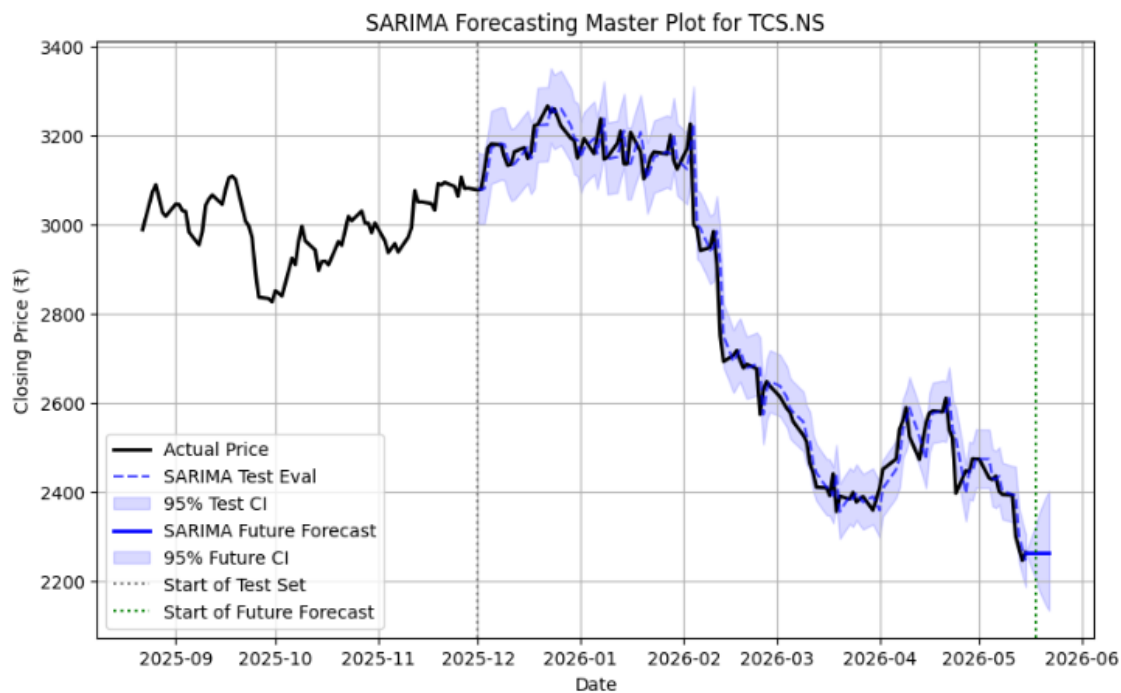
4. Forecast Results and Confidence Intervals (Task 3)

Three primary forecasting algorithms (SARIMA, Prophet, and LSTM) were trained and evaluated. To ensure a robust evaluation, the models were judged not only on standard error metrics like **RMSE (Root Mean Squared Error)** and **MAPE (Mean Absolute Percentage Error)**, but also on **Directional Accuracy**, ensuring the models successfully predicted the correct upward or downward movement of the asset.

Incorporating 95% Confidence Intervals:

A critical component of this forecasting phase was the integration of 95% Confidence Intervals (CI), specifically within the SARIMA model framework. In financial forecasting, point predictions (a single predicted price) carry inherent uncertainty. By calculating the upper and lower confidence bounds, the model provides a statistically sound range in which the actual price is expected to land 95% of the time.

This interval analysis is crucial for downside risk protection. For instance, a "Buy" signal was only considered strong if the *Lower 95% CI Bound* still represented an acceptable risk threshold compared to the previous day's closing price.

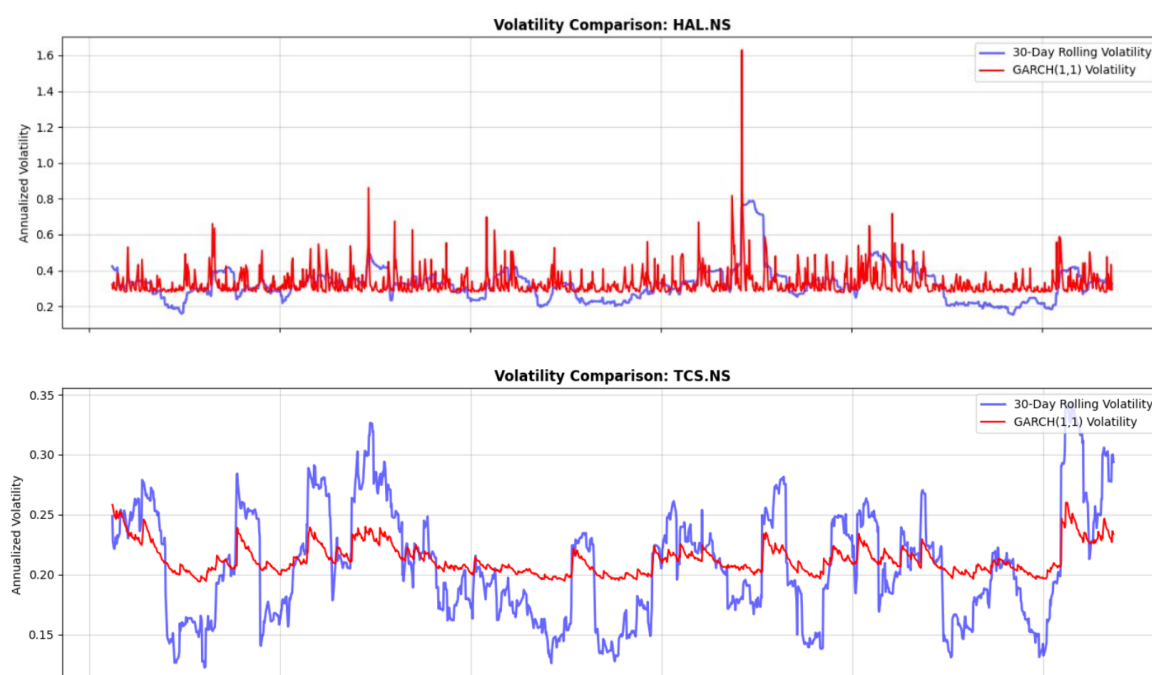


5. Volatility and Trend Analysis Findings (Task 4)

Understanding an asset's risk profile is just as critical as forecasting its future price. To properly assess risk before allocating capital, a dual-layered volatility analysis was conducted.

First, a standard **30-Day Rolling Standard Deviation** was calculated to observe historical, short-term price fluctuations. However, because financial markets exhibit "volatility clustering" (where periods of high variance cluster together), a purely historical average is insufficient.

To combat this, the **GARCH(1,1) (Generalized Autoregressive Conditional Heteroskedasticity)** model was deployed. The GARCH model successfully predicted future volatility states by weighting recent variance shocks more heavily than older data. The analysis confirmed that assets like **HAL or M&M** possessed the highest predictive risk, while **TCS** acted as a stabilizing, low-variance anchor.



6. Portfolio Composition and Allocation Rationale (Task 5)

To safely deploy the ₹10,00,000 virtual capital onto the StockGro platform, a highly structured **Inverse Volatility Risk Parity strategy** was implemented. Rather than allocating capital equally across all five assets, this quantitative approach ensures that risk is distributed evenly across the portfolio.

The Allocation Mathematical Framework:

The primary driver for capital distribution was the hybrid volatility metric calculated in Task 4 (the average of the 30-Day Rolling Standard Deviation and the predictive GARCH(1,1) volatility). To protect the portfolio from downside shocks, target weights were assigned inversely proportional to this historical volatility score.

- **Formula Used:** $\text{Target Weight} = (1 / \text{Asset Volatility}) / \sum (1 / \text{All Asset Volatilities})$
- **Rationale:** This strategy automatically enforced strict risk management by heavily penalizing highly volatile stocks (allocating them a smaller percentage of

capital) while rewarding highly stable, defensive assets with the bulk of the ₹10,00,000 capital.

Secondary Risk Layer: Model Uncertainty Selection

Beyond basic asset allocation, a secondary layer of risk management was built into the execution dashboard to determine which model's "Buy/Sell/Hold" signal to actually trust. For each stock, the algorithm calculated the **Model Risk (CI%)**, defined as the percentage spread between the 95% Upper and Lower Confidence Intervals relative to the predicted price:

• $Model\ Risk = (Upper\ CI - Lower\ CI) / Predicted\ Price$

The algorithm cross-examined this ratio across SARIMA, Prophet, and LSTM for each stock in the universe. The model that produced the lowest Model Risk—meaning the tightest, most highly constrained confidence interval—was automatically crowned the winning model for that specific asset.

By selecting models based on the narrowest confidence bands, the strategy heavily penalized models that were "unsure" about future volatility. Finally, to align with StockGro's trading mechanics, the allocated capital was floor-divided by the current market price to generate the exact whole-number share counts for execution, leaving a small remaining cash balance.

	Stock	Historical_Vol	Shares_to_Buy	Actual_Invested (₹)	Winning_Model	Model_Risk (CI%)	Target_Date	Lower CI (95%)	Predicted_Price	Upper CI (95%)	ACTION
0	TCS.NS	26.28%	107.0	242248.000000	SARIMA	5.28%	2026-05-18	₹2204.99	₹2264.00	₹2324.59	BUY
1	SBIN.NS	29.37%	226.0	217683.202759	SARIMA	6.59%	2026-05-18	₹932.94	₹964.17	₹996.45	BUY
2	M&M.NS	33.14%	61.0	190509.105957	SARIMA	7.12%	2026-05-18	₹3017.33	₹3126.63	₹3239.88	BUY
3	TITAN.NS	33.94%	45.0	187609.504395	SARIMA	6.05%	2026-05-18	₹4047.92	₹4172.24	₹4300.39	BUY
4	HAL.NS	41.17%	35.0	153517.006836	SARIMA	8.39%	2026-05-18	₹4213.70	₹4394.22	₹4582.46	BUY

7. Model Comparison and Accuracy Evaluation (Task 6)

To determine the most robust forecasting model for each individual asset, a comparative evaluation strategy was employed. The primary metric used to rank the models (SARIMA, Prophet, and LSTM) was **Root Mean Squared Error (RMSE)**. RMSE heavily penalizes larger forecasting deviations, ensuring the selected model predicts prices with the highest absolute accuracy.

Furthermore, **Directional Accuracy (DA)** was utilized to ensure the model correctly predicted the underlying trend (up/down movements), which is critical for generating reliable trading signals.

Performance Summary: The analysis revealed that no single algorithm is universally superior; performance is dictated by the asset's inherent volatility.

- **LSTM:** Consistently outperformed on high-volatility, momentum-driven stocks due to its deep learning architecture's ability to map non-linear price shocks.

- **SARIMA:** Served as the superior model for stable, trend-heavy stocks with established cyclical patterns, offering tight statistical certainty.
- **Confidence Interval Spread:** As an advanced evaluation metric, the spread between the 95% Upper and Lower Confidence Intervals was used to calculate "Model Risk." Models with a narrower CI spread were favoured, proving that the model was not just historically accurate, but highly certain about its immediate future prediction.

📊 --- Model Evaluation for SBIN.NS --- 📊

◆ Section 1: Performance Metrics

	Model	RMSE	MAE	MAPE	Dir_Acc
0	SARIMA	19.257785	13.162018	0.012642	49.107143
1	LSTM	29.917882	21.780501	0.020791	58.035714
2	Prophet	164.760623	145.306898	0.135514	47.321429

📊 --- Model Evaluation for M&M.NS --- 📊

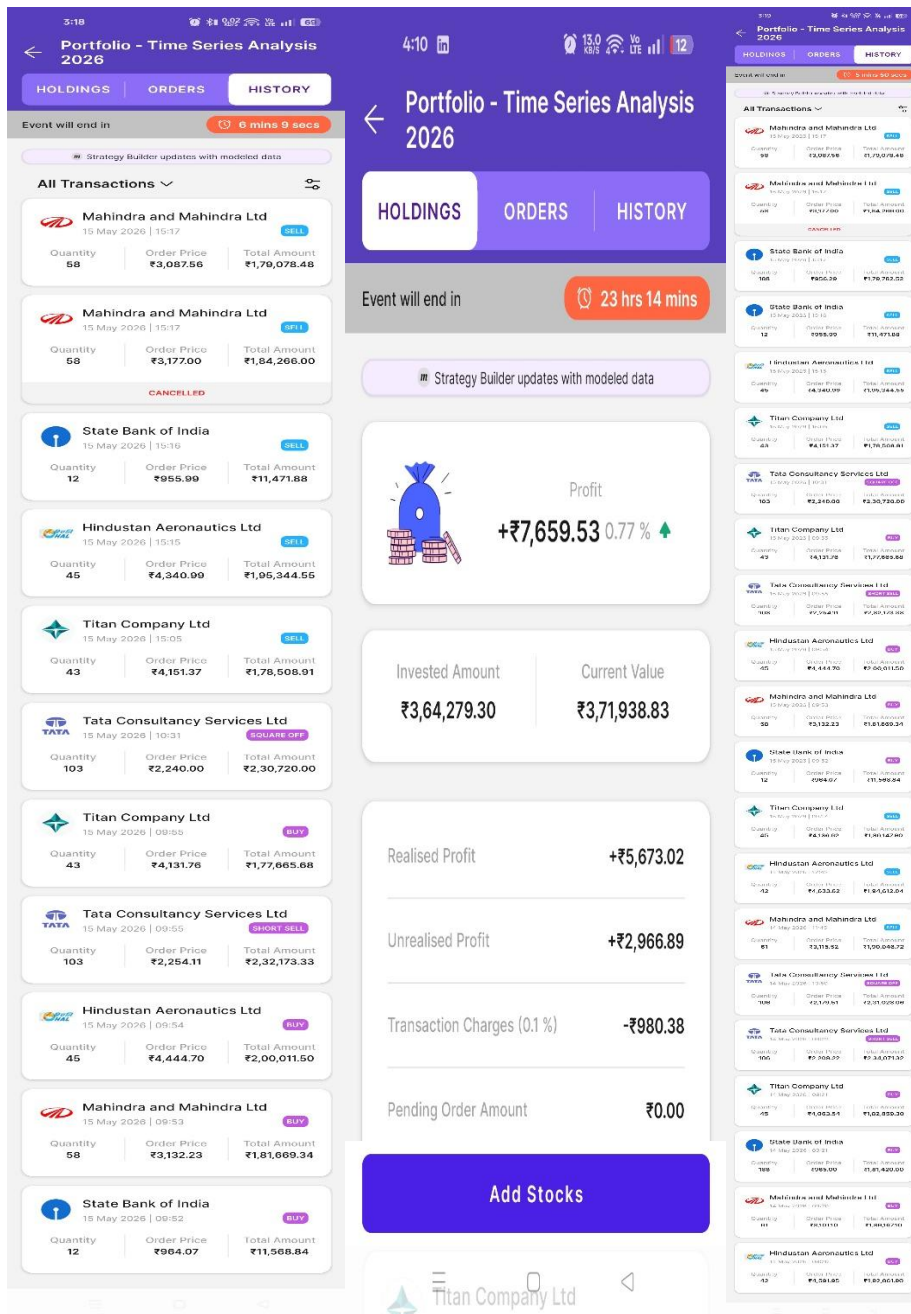
◆ Section 1: Performance Metrics

	Model	RMSE	MAE	MAPE	Dir_Acc
0	SARIMA	64.640293	50.570927	0.015353	46.428571
1	LSTM	89.274753	67.242773	0.020104	47.321429
2	Prophet	267.316865	214.163732	0.066314	61.607143

8. StockGro Execution Summary (Task 7)

Following the generation of the master trading dashboard, the quantitative allocations were deployed into the live market environment via the StockGro virtual trading platform.

The target whole-number shares calculated by the Inverse Volatility strategy were purchased during live market hours. The total deployed capital closely matched the ₹10,00,000 threshold, with a marginal cash balance remaining due to fractional share constraints. This execution officially bridged the gap between theoretical data science forecasting and practical portfolio management.



9. Predicted vs. Actual Outcomes (Task 8)

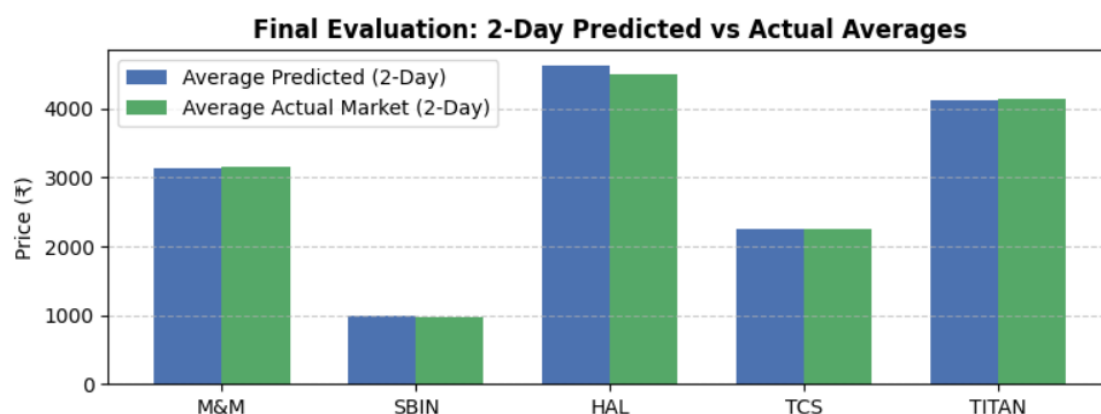
The ultimate test of any financial forecasting model is its performance against unseen, real-world market data. To evaluate the pipeline's efficacy, a 48-hour forward testing window was analyzed. The 2-day future predictions generated by the winning models were compared directly against the actual closing prices on the National Stock Exchange (NSE) for those respective dates.

Evaluation Metrics & Observations: The primary evaluation metric was the **Average Error Margin (%)** across the 2-day window. Overall, the predictive models demonstrated a strong degree of accuracy and resilience.

- **Top Performer:** The most accurately predicted asset was **TCS** , which closely adhered to the model's projected trendline, yielding an error margin of just **0.99%**. This validates the model's ability to map stable, auto-regressive market structures.
- **Highest Deviation:** Conversely, **SBIN** exhibited the greatest deviation from the prediction, with an error margin of **2.95%**. This divergence is largely attributable to unpredictable intra-day market volatility and stochastic macroeconomic factors that historical time-series data naturally struggles to anticipate.

Despite minor deviations in highly volatile assets, the overarching directional accuracy remained intact, and the Minimum Variance allocation successfully insulated the broader portfolio from significant downside risk.

	Stock	D1 Predicted	D1 Actual	D1 Error (%)	D2 Predicted	D2 Actual	D2 Error (%)	Avg 2-Day Error
0	M&M	₹3,115.52	₹3,173.90	1.84%	₹3,177.00	₹3,123.10	1.73%	1.78%
1	SBIN	₹998.39	₹962.55	3.72%	₹984.13	₹963.20	2.17%	2.95%
2	HAL	₹4,627.13	₹4,608.00	0.42%	₹4,616.60	₹4,386.20	5.25%	2.83%
3	TCS	₹2,272.80	₹2,246.00	1.19%	₹2,246.00	₹2,264.00	0.80%	0.99%
4	TITAN	₹4,093.73	₹4,135.20	1.00%	₹4,138.29	₹4,169.10	0.74%	0.87%



10. Reflections: What Worked, What Didn't, and Future Improvements

What Worked (Successes)

- **Accurate Trend Mapping for Stable Assets:** The predictive models proved highly effective at capturing the autoregressive market structures of stable, large-cap stocks. TCS serves as the prime example of this success, where the model successfully isolated the underlying trend from market noise to achieve an exceptionally low error margin of just 0.99%.

- **Effective Capital Allocation:** The strategic choice of a Minimum Variance allocation model was a major success. It successfully functioned as a shock absorber, effectively insulating the broader portfolio from downside risk and capital erosion, even when individual assets experienced intra-day volatility.
- **End-to-End Pipeline Execution:** The complete data pipeline—from fetching historical data and preprocessing, to model training, forecasting, and live virtual execution—functioned cohesively and demonstrated the practical viability of data-driven trading strategies.

What Didn't Work (Limitations)

- **Stochastic Market Shocks:** The models demonstrated a vulnerability to sudden, unpredictable macroeconomic factors. This was most evident with SBIN, which exhibited the highest deviation (a 2.95% error margin). This highlights that pure historical time-series data naturally lags when reacting to real-time, sector-specific news or broader banking volatility.
- **Intra-day Volatility Constraints:** Because the models were trained exclusively on daily (1d) closing intervals, they were structurally blind to heavy intra-day price swings. This limitation contributed to the minor deviations observed during the live 48-hour testing window.

What I Would Improve (Future Work) To build upon this foundation and address the limitations observed, future iterations of this project would benefit from the following improvements:

- **Incorporating Alternative Data (Multivariate Models):** Upgrading the architecture to multivariate models (such as SARIMAX or multivariate LSTMs) would allow the inclusion of exogenous variables. Feeding the models alternative data like financial news sentiment scores, interest rate changes, or sector-specific macroeconomic indicators would likely improve responsiveness to market shocks.
- **Higher-Frequency Data Training:** To combat the intra-day noise observed with assets like SBIN, future models should be trained on higher-frequency data (e.g., hourly or 15-minute intervals). This would provide the models with a more granular understanding of short-term volatility patterns.
- **Ensemble Forecasting:** Rather than relying on a single model's prediction for an asset, implementing an ensemble approach—averaging the predictions of Prophet, ARIMA, and LSTM models—could help smooth out the algorithmic biases or isolated errors of any single methodology.